COMPARISON TEST FOR IMPROPER INTEGRALS

LEARNING OBJECTIVES

Upon completion of this unit, you should be able to:

- Identify and determine the convergence or divergence of improper integral of type I.
- Identify and determine the convergence or divergence of improper integral of type II.
- Use comparison test to determine the convergence or divergence of improper integrals.

INTRODUCTION

Sometimes it is difficult to find the exact value of an improper integral by antidifferentiation. However, it is still possible to determine whether an improper integral converges or diverges. The idea is to compare the integral to one whose behavior we already know.

IMPROPER INTEGRALS

The definite integral $\int_a^b f(x)dx$ is called an improper integral if

1. At least one of the limits of integration is infinite, or
2. The integrand $f(x)$ has one or more points of discontinuity on the interval $[a, b]$.

TYPE I IMPROPER INTEGRALS

1. If $\int_a^t f(x)dx$ exists for every number $t \geq a$, then $\int_a^\infty f(x)dx = \lim_{t\to\infty} \int_a^t f(x)dx$, provided this limit exists.
2. If $\int_t^b f(x)dx$ exists for every number $t \leq b$, then $\int_{-\infty}^b f(x)dx = \lim_{t\to-\infty} \int_t^b f(x)dx$, provided this limit exists.

The improper integrals $\int_a^\infty f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called convergent if the corresponding limit exists and divergent if the limit does not exist.

3. If $\int_a^\infty f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are convergent, then $\int_{-\infty}^\infty f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^\infty f(x)dx$.

TYPE I IMPROPER INTEGRALS

Example 1.

Determine whether $\int_0^{\infty} xe^{-x}dx$ is convergent or divergent.

Solution

$$\int_0^{\infty} xe^{-x}dx = \lim_{t\to\infty}\int_0^{t} xe^{-x}dx = \lim_{t\to\infty} -xe^{-x} - e^{-x}\Big|_0^t$$

$$= \lim_{t\to\infty}[-te^{-t} - e^{-t} + 0 + e^{-0}] = \lim_{t\to\infty}[-te^{-t} - e^{-t} + 1].$$

Notice that $\lim_{t\to\infty}[-te^{-t}] \longrightarrow \frac{-\infty}{\infty}$. Therefore, applying L'Hopital Rule, we have $\lim_{t\to\infty}[-te^{-t}] = \lim_{t\to\infty}\left[\frac{-t}{e^t}\right] = \lim_{t\to\infty}\left[\frac{-1}{e^t}\right] = 0$, so that

$$\int_0^{\infty} xe^{-x}dx = \lim_{t\to\infty}[-te^{-t} - e^{-t} + 1] = (0 + 0 + 1) = 1.$$

Therefore, $\int_0^{\infty} xe^{-x}dx$ converges to 1.

TYPE I IMPROPER INTEGRALS

Example 2.

Determine whether $\int_{-\infty}^{\infty}(2x^2-x+3)dx$ is convergent or divergent.

Solution

$$\int_{-\infty}^{\infty}(2x^2-x+3)dx=\int_{-\infty}^{0}(2x^2-x+3)dx+\int_{0}^{\infty}(2x^2-x+3)dx$$

$$=\lim_{a\to-\infty}\int_{a}^{0}(2x^2-x+3)dx+\lim_{b\to\infty}\int_{0}^{b}(2x^2-x+3)dx$$

$$=\lim_{a\to-\infty}\left[\frac{2x^3}{3}-\frac{x^2}{2}+3x\right]\Bigg|_{a}^{0}+\lim_{b\to\infty}\left[\frac{2x^3}{3}-\frac{x^2}{2}+3x\right]\Bigg|_{0}^{b}=\infty$$

Therefore, $\int_{-\infty}^{\infty}(2x^2-x+3)dx$ diverges.

TYPE II IMPROPER INTEGRALS

1. If f is continuous on the interval $[a, b)$ and has an infinite discontinuity at b, then $\int_a^b f(x)dx = \lim_{c \to b^-} \int_a^c f(x)dx$.
2. If f is continuous on the interval $(a, b]$ and has an infinite discontinuity at a, then $\int_a^b f(x)dx = \lim_{c \to a^+} \int_c^b f(x)dx$.
3. If f is continuous on the interval $[a, b]$ except for some c in (a, b) at which f has an infinite discontinuity, then $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$.

TYPE II IMPROPER INTEGRALS

Example 1.

Determine whether $\int_{-1}^{2} \frac{1}{x^4}\, dx$ is convergent or divergent.

Solution

The function is undefined at $x = 0$. Therefore, we could define this integral as

$$\int_{-1}^{2} \frac{1}{x^4}\, dx = \int_{-1}^{0} \frac{1}{x^4}\, dx + \int_{0}^{2} \frac{1}{x^4}\, dx$$

$$\int_{-1}^{2} \frac{1}{x^4}\, dx = \lim_{c \to 0^-} \int_{-1}^{c} \frac{1}{x^4}\, dx + \lim_{b \to 0^+} \int_{b}^{2} \frac{1}{x^4}\, dx$$

$$= \lim_{c \to 0^-} \left[\frac{-1}{3x^3}\right]\Bigg|_{-1}^{c} + \lim_{b \to 0^+} \left[\frac{-1}{3x^3}\right]\Bigg|_{b}^{2}$$

Notice that $\left[\frac{-1}{3x^3}\right] \longrightarrow -\infty$ as $c \to 0^-$ and $\left[\frac{-1}{3x^3}\right] \longrightarrow -\infty$ as $b \to 0^+$. Therefore, $\int_{-1}^{2} \frac{1}{x^4}\, dx$ is divergent.

COMPARISON TEST FOR IMPROPER INTEGRALS

Theorem. Let $f(x)$ and $g(x)$ be any two continuous functions such that $0 \leq g(x) \leq f(x)$ for all $x \geq a$. Then

1. If $\int_a^\infty f(x)dx$ is convergent, then $\int_a^\infty g(x)dx$ is convergent,
2. If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent.

COMPARISON TEST FOR IMPROPER INTEGRALS

Example 1.

Determine whether $\int_2^\infty \frac{cos^2(x)}{x^2}\,dx$ converges or diverges.

Solution

For all $x \geq 2$, it can be seen that since $0 \leq cos^2(x) \leq 1$, then $\frac{cos^2(x)}{x^2} \leq \frac{1}{x^2}$. Let $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{cos^2(x)}{x^2}$. Then $0 \leq g(x) \leq f(x)$.

Now,

$$\int_2^\infty f(x)dx = \int_2^\infty \frac{1}{x^2}dx = \lim_{x\to\infty}\int_2^\infty \frac{1}{x^2}dx = \left(\frac{-1}{\infty} - \frac{(-1)}{2}\right) = \frac{1}{2}$$

Since $\int_2^\infty \frac{1}{x^2}dx$ is convergent, then by the comparison test, $\int_2^\infty \frac{cos^2(x)}{x^2}dx$ is also convergent.

Note 1. The integral $\int_1^\infty \frac{1}{x^p}dx$ converges if $p > 1$ and diverges if $p \leq 1$.

COMPARISON TEST FOR IMPROPER INTEGRALS

Example 2.

Determine whether $\int_3^\infty \frac{1}{\sqrt{x^2-1}}\,dx$ converges or diverges.

Solution

For all $x \geq 3$, it can be seen that $\frac{1}{\sqrt{x^2-1}} \geq \frac{1}{x}$. By the p-integral test, $\int_3^\infty \frac{1}{x}\,dx$ diverges by the above Note 1, since $p = 1$. Therefore, by the comparison test, $\int_3^\infty \frac{1}{\sqrt{x^2-1}}\,dx$ is divergent.

COMPARISON TEST FOR IMPROPER INTEGRALS

Example 3.

Determine whether $\int_1^\infty \frac{1}{\sqrt{x^3+7}}dx$ converges or diverges.

Solution

For all $x \geq 1$, it can be seen that $x^3 + 7 \geq x^3$, and $\sqrt{x^3+7} \geq \sqrt{x^3}$

implying that $\frac{1}{\sqrt{x^3+7}} \leq \frac{1}{\sqrt{x^3}}$. Let $f(x) = \frac{1}{\sqrt{x^3}}$ and $g(x) = \frac{1}{\sqrt{x^3+7}}$. Then $0 \leq g(x) \leq f(x)$.

Now, $\int_1^\infty f(x)dx = \int_1^\infty \frac{1}{\sqrt{x^3}}dx = \int_1^\infty \frac{1}{x^{\frac{3}{2}}}dx$. By the above Note 1, $p = \frac{3}{2} > 1$. Therefore, $\int_1^\infty \frac{1}{x^{\frac{3}{2}}}dx$ is convergent. By the comparison test, $\int_1^\infty \frac{1}{\sqrt{x^3+7}}dx$ is convergent.

TASK

Determine whether $\int_1^\infty \frac{1}{\sqrt{x^6+1}}\,dx$ converges or diverges.

SUMMARY

The definite integral is called an improper integral if at least one of its limits is infinite, or the integrand has one or more points of discontinuity on the interval of definition.

If $f(x)$ and $g(x)$ are any two continuous functions such that $0 \leq g(x) \leq f(x)$ for all $x \geq a$. Then if $\int_a^{\infty} f(x)dx$ is convergent, then so is $\int_a^{\infty} g(x)dx$. If $\int_a^{\infty} g(x)dx$ is divergent, then so is $\int_a^{\infty} f(x)dx$.

The integral $\int_1^{\infty} \frac{1}{x^p} dx$ converges if $p > 1$ and diverges if $p \leq 1$.

FURTHER READING

"Calculus" by James Stewart.

"Calculus: Early Transcendentals" by Howard Anton, Irl Bivens, and Stephen Davis.

" Calculus Know-It-ALL " by Stan Gibilisco.

" Theory and Problems of ADVANCED CALCULUS" by Robert Wrede, and Murray R. Spiegel.

“Advanced Engineering Mathematics" by H. K. Dass.

“Engineering Mathematics" by K. A. Stroud.

THANK YOU